

It is a 4-Layer Model comprises of

- 1. Application Layer:**
 - 2. Transport Layer:**
 - 3. Network Layer:**
 - 4. Physical Layer:**
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Explain DHCP DORA

DHCP is a Dynamic Host Configuration Protocol, which automatically assigns IP addresses and other network settings to devices on a network. DORA is an acronym that represents the four-step process DHCP uses to assign these IP addresses: Discover, Offer, Request, Acknowledge. PORT NUMBERS – UDP 67 for server, UDP 68 for the Client.

Reverse Proxy

A reverse proxy acts as an intermediary server sitting in front of one or more web servers. It receives client requests, distributes them to the appropriate backend server based on rules, and forwards the response back to the client. The client perceives the reverse proxy as the origin server.

NAT, which stands for Network Address Translation, is a technique used in computer networks to manage how devices connect to the internet. It essentially acts as a translator between private IP addresses used within a local network and public IP addresses used on the wider internet.

HTTP REQUEST METHODS

Here are some of the most common HTTP request methods:

- GET:
- POST:
- PUT:
- DELETE:

- PATCH:
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Types of DNS Records:

A Record: Maps a domain name (e.g., "www.example.com") to an IPv4 address (e.g., "192.168.1.1"). This is the most fundamental DNS record type.

AAAA Record: Maps a domain name to an IPv6 address. IPv6 is the next generation of IP addressing due to the exhaustion of IPv4 addresses.

MX Record: Specifies mail exchange servers responsible for handling email delivery for a domain.

CNAME Record: Creates an alias for another domain name. This is useful for subdomains (e.g., "[invalid URL removed]" can be an alias for "[invalid URL removed]").

TXT Record: Stores arbitrary text data associated with a domain name. It can be used for various purposes like email authentication or website verification.

Here's a breakdown of the different types of Active Directory partitions:

1. Domain Partition (Naming Context):
 2. Schema Partition (Naming Context):
 3. Configuration Partition (Naming Context):
 4. Application Partition (Optional):
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The commands to check TCP/IP configuration differ depending on your operating system:

ipconfig: Shows basic information like IP address, subnet mask, default gateway, and DNS server for all network adapters.

ipconfig /all: Provides a more detailed view, including hostname, MAC address, DHCP lease information, etc.

ping: This command is used to test connectivity to a specific IP address or hostname. It sends packets to the target and measures the response time, helping diagnose network issues.

List FSMO Roles

There are five Flexible Single Master Operation (FSMO) roles in Active Directory:

Schema Master: There is only one schema master per forest. It holds the schema, which defines the structure of all objects stored in Active Directory, such as users, groups, and computers. Any changes to the schema must be made on the schema master DC and then replicated to all other domain controllers in the forest.

Domain Naming Master (DNM): There is only one DNM per forest. It manages the creation and deletion of domains within the forest. Additionally, it stores forest-wide configuration data like domain trusts and group policy objects (GPOs) applied at the forest level.

Relative ID (RID) Master: Each domain has its own RID master. It's responsible for issuing unique security identifiers (SIDs) to objects within the domain. SIDs are essential for user and group permissions across the domain.

Primary Domain Controller (PDC) Emulator: Each domain has its own PDC emulator. It provides compatibility features for applications that rely on older domain controllers and performs certain time synchronization tasks within the domain.

Infrastructure Master: Each domain has its own infrastructure master. It keeps track of updates made to objects in the domain and ensures proper replication of these changes to other domain controllers.

Distinguish between a process and a thread

Process:

A process is an instance of a program in execution. It's the basic unit of work allocation in an operating system.

A process has its own independent set of resources, including memory space, program code, CPU registers, and open files. These resources are isolated from other processes.

Example: Opening a new application like a web browser or a game creates a new process.

Thread:

A thread is a lightweight unit of execution within a process. It shares the same address space and resources (memory, code, files) with the main process and other threads within the process.

Threads do not have their own independent resources. They rely on the resources allocated to the parent process.

Example: A web browser might create multiple threads to handle different tasks like loading images, processing user input, and managing network connections. These threads work together within the single web browser process.

What is a child DC

In the context of Active Directory, a child domain controller (DC) is a domain controller that replicates data from a parent domain controller within the same domain.

Here's an analogy to understand child DCs:

Imagine a company with a main office (parent domain) and branch offices (child domains). The main office has all the company's central records. Branch offices need copies of some of this data (like employee information) but don't need the entire dataset. Child domain controllers act like file servers in the branch offices, replicating relevant data from the main office (parent domain controller).

SYSVOL, which stands for System Volume, is a critical directory in Active Directory (AD) that resides on every domain controller (DC) within a domain. It acts as a central repository for files that need to be replicated and accessed consistently across all domain controllers.

CSR

Certificate Signing Request: This is the most common meaning of CSR. It refers to a message sent from an applicant (like a website owner) to a certificate authority (CA) to request a digital security certificate.

Purpose: An SSL/TLS certificate establishes a secure connection between a web server and a client (web browser). A CSR is the first step in obtaining an SSL/TLS certificate. It contains information about the applicant's organization and public key, which will be used to create the certificate.

A CSR typically includes the following information:

Common Name (CN): The domain name for which the certificate is being requested (e.g., "www.example.com").

Organization Name (O): The legal name of the organization requesting the certificate.

Locality (L): The city or town where the organization is located.

State or Province Name (ST): The state or province where the organization is located.

Country Name (C): The two-letter country code (e.g., "US" for United States).

Public Key: The public key of the applicant's key pair. This key pair is used for encryption and decryption in the secure connection.

SAML AUTHENTICATION

SAML stands for Security Assertion Markup Language. It's an industry-standard protocol for authentication and authorization between entities in a web-based environment. Imagine SAML as a secure way for different websites to exchange information about your identity, allowing you to log in once and access multiple applications without needing to re-enter credentials everywhere.

Here's a breakdown of how SAML authentication works:

The Players involved:

Identity Provider (IdP): This is a trusted service that holds user credentials and performs the initial user authentication. Often, this could be your company's login portal or a cloud-based identity provider service.

Service Provider (SP): This is the website or application that you want to access using SAML authentication. Examples include web applications, cloud storage services, or corporate portals.

User: You, the end-user trying to access the service provider's application.

The Authentication Flow:

User tries to access a Service Provider (SP): You attempt to access a web application that uses SAML authentication.

Redirect to Identity Provider (IdP): The SP recognizes SAML and redirects you to the login page of your IdP. This is where you enter your username and password.

User Authentication at IdP: The IdP authenticates you using your username and password. If successful, the IdP creates a SAML assertion containing information about your identity and access rights.

SAML Assertion sent to SP: The IdP sends the SAML assertion back to the Service Provider. This assertion is a digitally signed token that securely confirms your identity and relevant access rights.

User Access Granted (or Denied): The SP receives the SAML assertion, validates it with the IdP using digital signatures, and grants you access to the application based on the information in the assertion.

Kerberos, LDAP, HTTP, HTTPS PORT NUMBERS

Kerberos:

UDP/TCP port 88 - This is the default port used by Kerberos for communication between clients and the Kerberos Key Distribution Center (KDC).

LDAP:

TCP port 389 - This is the standard port used for unencrypted LDAP communication.

TCP port 636 - This is the port used for LDAP communication secured with SSL/TLS, often referred to as LDAPS.

HTTP:

TCP port 80 - This is the well-known port for standard HTTP communication.

HTTPS:

TCP port 443 - This is the universally recognized port for HTTPS communication, which uses SSL/TLS for encryption.

LSDOU stands for Local Site Domain Organizational Unit. It represents the order in which Group Policy Objects (GPOs) are applied in a Windows Active Directory

environment. Here's a breakdown of what each element represents and how they influence GPO application:

Order of Precedence:

LSDOU defines a hierarchical order for processing GPOs. Policies applied at higher levels in the hierarchy override conflicting settings from lower levels. This ensures granular control over GPO application within your Active Directory domain.

Breaking Down LSDOU:

1. **Local:**
 - GPOs defined at the local computer level. These policies apply specifically to the machine they reside on and have the highest precedence.
 2. **Site:**
 - GPOs linked to a specific Active Directory site. Site-level GPOs apply to all domain controllers and member computers within that site.
 3. **Domain:**
 - GPOs linked to the domain itself. Domain-level GPOs apply to all domain controllers and member computers across the entire domain, regardless of site location.
 4. **Organizational Unit (OU):**
 - GPOs linked to an OU within the Active Directory domain structure. OU-level GPOs apply to all member computers and users contained within that specific OU and its child OUs. GPOs applied at child OUs override conflicting settings from parent OUs.
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